



IDC 401: Theoretical Biology

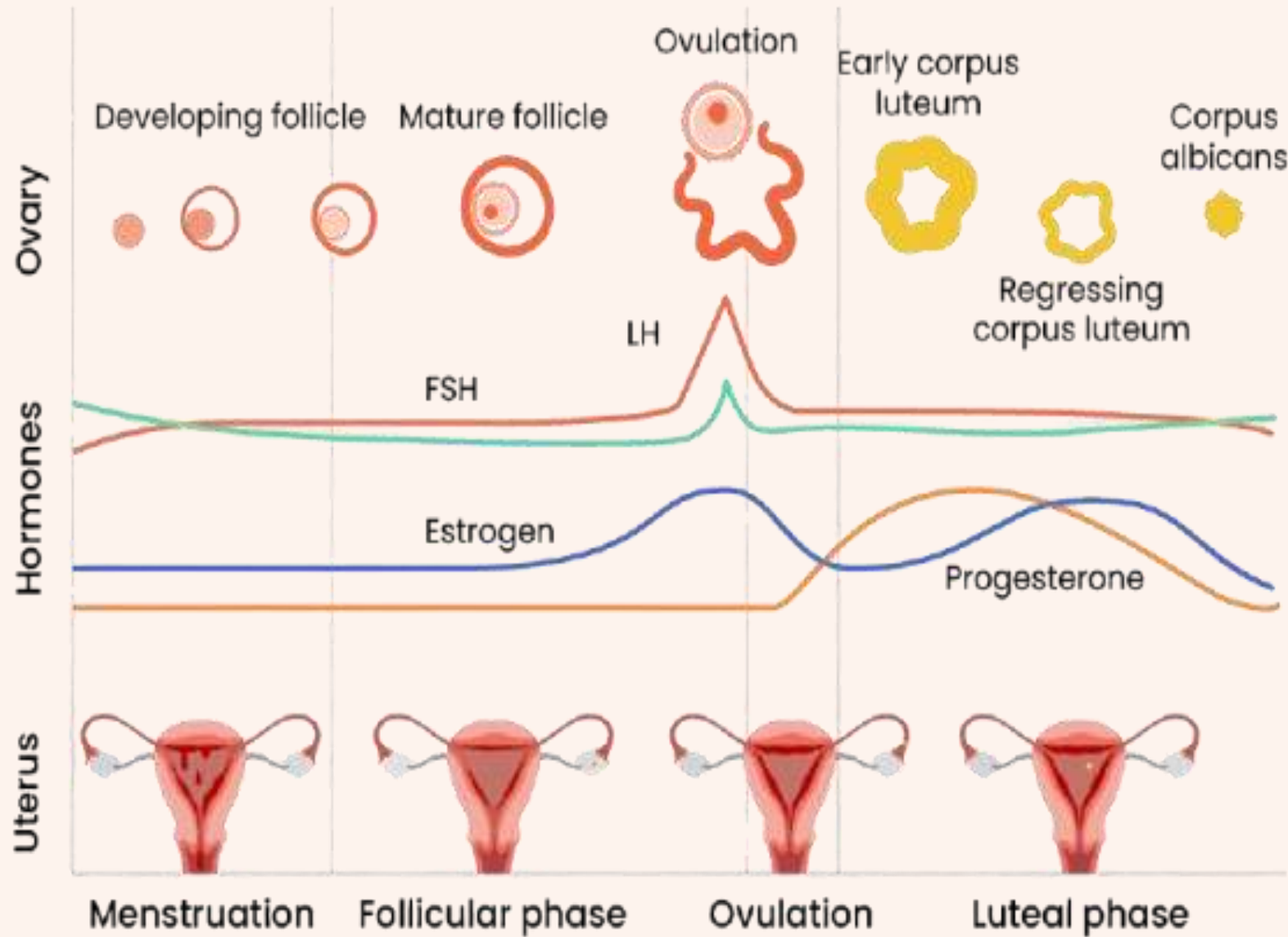
# Diagnosing Endometriosis using Hormone Level Modelling

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# INTRODUCTION



**Proliferative Phase**  
↓  
**Follicles maturation**  
↓  
**Estrogen↑**  
↓  
**Endometrium thickens**

**Ovulation**  
↓  
**Progesterone↑**  
↓  
**Secretory Phase**  
↓  
**Menstruation**

# INTRODUCTION

- When the endometrium grows excessively, it could initiate ectopic growth, known as endometriosis.
- It causes pain, menstrual irregularities, bladder dysfunction and infertility risks.
- Caused by a multifactorial condition due to which its etiology is still not understood well.
- **It is known that excessive endometrial growth occurs in response to the presence of estrogen, but it doesn't imply that estrogen is the root cause of this growth.**
- Studies have now shown that a healthy female reproductive system including a healthy endometrial growth is a balance between estrogen and progesterone.

# Key Literature

- **Coupled Model simulations:**

*Carolina Ramirez Mazo, Laura Lema-Perez, Lina Gomez Echavarria* “Endometriosis: A parametric analysis of a phenomenological-based endometrial model” *2021 IEEE 5th Colombian Conference on Automatic Control (CCAC), 2021.*

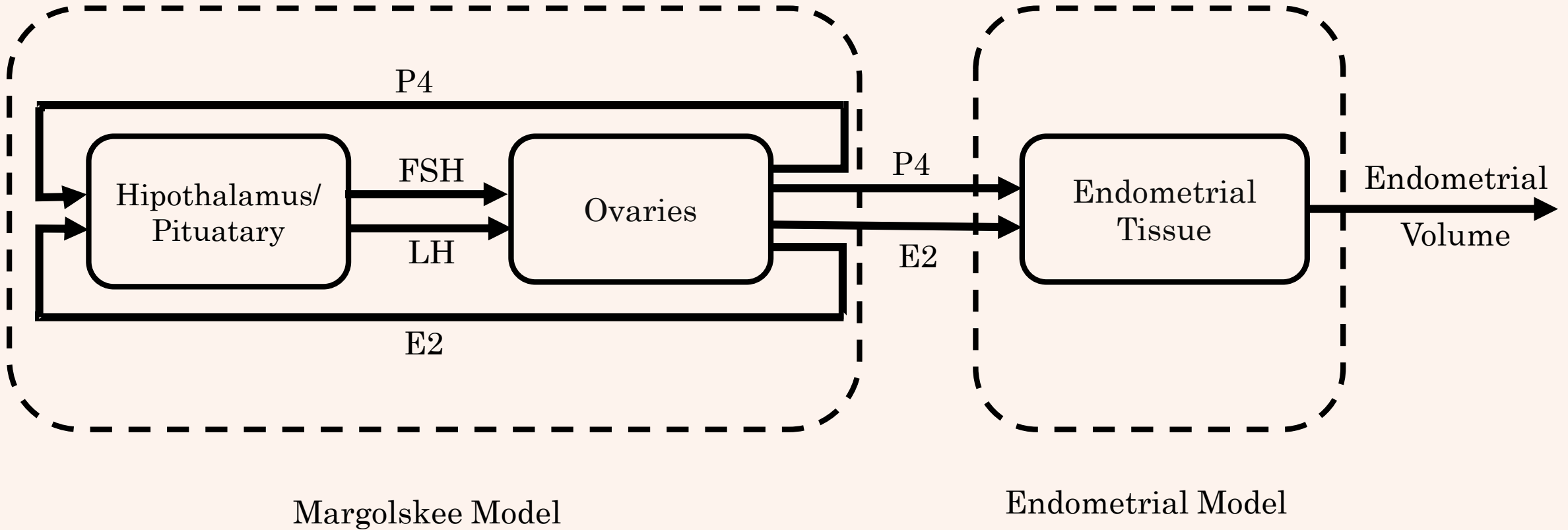
- **Hormone Model:**

*Alison Margolskee, James F. Selgrade* “A lifelong model for the female reproductive cycle with an antimüllerian hormone treatment to delay menopause” *Journal of Theoretical Biology* 326 (2013) 21-35.

- **Endometrial Model:**

*Daniela Arbeláez-Gómez, Santiago Benavides-López, Maria Paula Giraldo-Agudelo, Juan Pablo Guzmán-Álvarez, Carolina Ramirez-Mazo, Lina María Gómez-Echavarría* “A phenomenological-based model of the endometrial growth and shedding during the menstrual cycle” *Journal of Theoretical Biology* 532 (2022) 110922.

# Coupled Model



- It is a phenomenological model → Parametric interpretability → Improve decision making for devising applications.

# Assumptions & Symbols

- All endometrial cell types are indistinct. Only the phases of the cycle that affect them are differentiated.
- The law of conservation of mass is applied.
- Hormonal cycles are a 28 days periodic.
- Uterus is a homogeneous compartment.
- Each cell type has fixed mass density.

| Symbol                | Description                                                     | Value                         |
|-----------------------|-----------------------------------------------------------------|-------------------------------|
| $k_{pro}$             | Estrogen effect on the maximum growth of the cells              | $0.008417cm^3pg^{-1}day^{-1}$ |
| $k_{sec}$             | Progesterone effect on the maximum growth of the cells          | $0.154731cm^3pg^{-1}day^{-1}$ |
| $k_d$                 | Constant for the inverse effect of progesterone upon cell death | $20\ adim$                    |
| $K_S$                 | Substrate Constant                                              | $10\ mg$                      |
| $m_p$                 | Specific product formation rate                                 | $0.001day^{-1}$               |
| $M_S$                 | Mass of substrate inside the tissue                             | $1.17mg$                      |
| $Y_{x_p,x_{sec}}$     | Yield coefficient of product formation from proliferative cells | $0.1mg/mg$                    |
| $Y_{x_{pro},x_{sec}}$ | Yield coefficient of secretory from proliferative cells         | $0.45mg/mg$                   |
| $w_{x_{sec}}$         | Mass fraction of secretory cells                                | $0.9mg/mg$                    |
| $w_{x_p}$             | Mass fraction of product                                        | $0.1mg/mg$                    |
| $\rho_{x_{pro}}$      | Density of cells in proliferative phase                         | $1050mg/cm^3$                 |
| $\rho_{x_{sec}}$      | Density of cells in secretory phase                             | $1050mg/cm^3$                 |
| $\rho_{x_p}$          | Density of cellular product                                     | $1400mg/cm^3$                 |

# Equations

## Cell Proliferation

$$\mu_{max_{pro}} = k_{pro} u_{[E]}$$

$$\mu_{pro} = \mu_{max_{pro}} \frac{M_S}{K_S + M_S}$$

$$\frac{dx_{pro}}{dt} = \mu_{pro} x_{pro} - \mu_{sec} x_{pro}$$

## Cell Secretion

$$\mu_{max_{sec}} = k_{sec} u_{[P]}$$

$$\mu_{sec} = \mu_{max_{sec}} \frac{M_S}{K_S + M_S}$$

$$\frac{dx_{sec}}{dt} = Y_{x_{pro}, x_{sec}} \mu_{sec} x_{pro} - k_d S_B \left( \frac{x_{sec}}{w_{x_{sec}}} + \frac{x_p}{w_{x_p}} \right) w_{x_p}$$

## Product Formation

$$\frac{dx_p}{dt} = (Y_{x_p, x_{sec}} + \mu_p) x_{sec} - k_d S_B \left( \frac{x_{sec}}{w_{x_{sec}}} + \frac{x_p}{w_{x_p}} \right) w_{x_p}$$

## Total Endometrial Volume

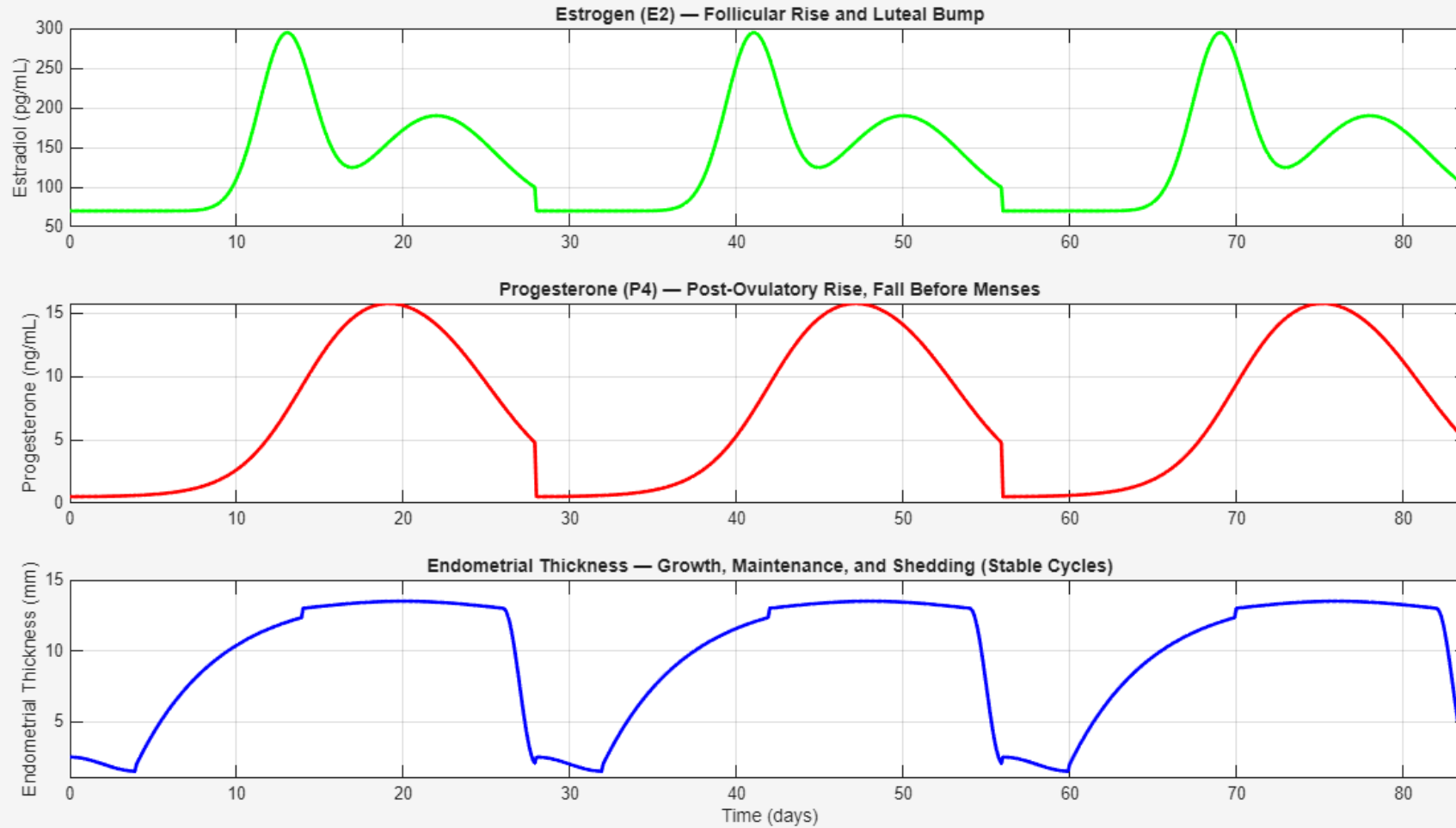
$$V_T = \frac{x_{pro}}{\rho_{x_{pro}}} + \frac{x_{sec}}{\rho_{x_{sec}}} + \frac{x_p}{\rho_{x_p}}$$

## E2-P4 Ratio

$$\frac{K_{pro}}{K_{sec}} = Constant$$

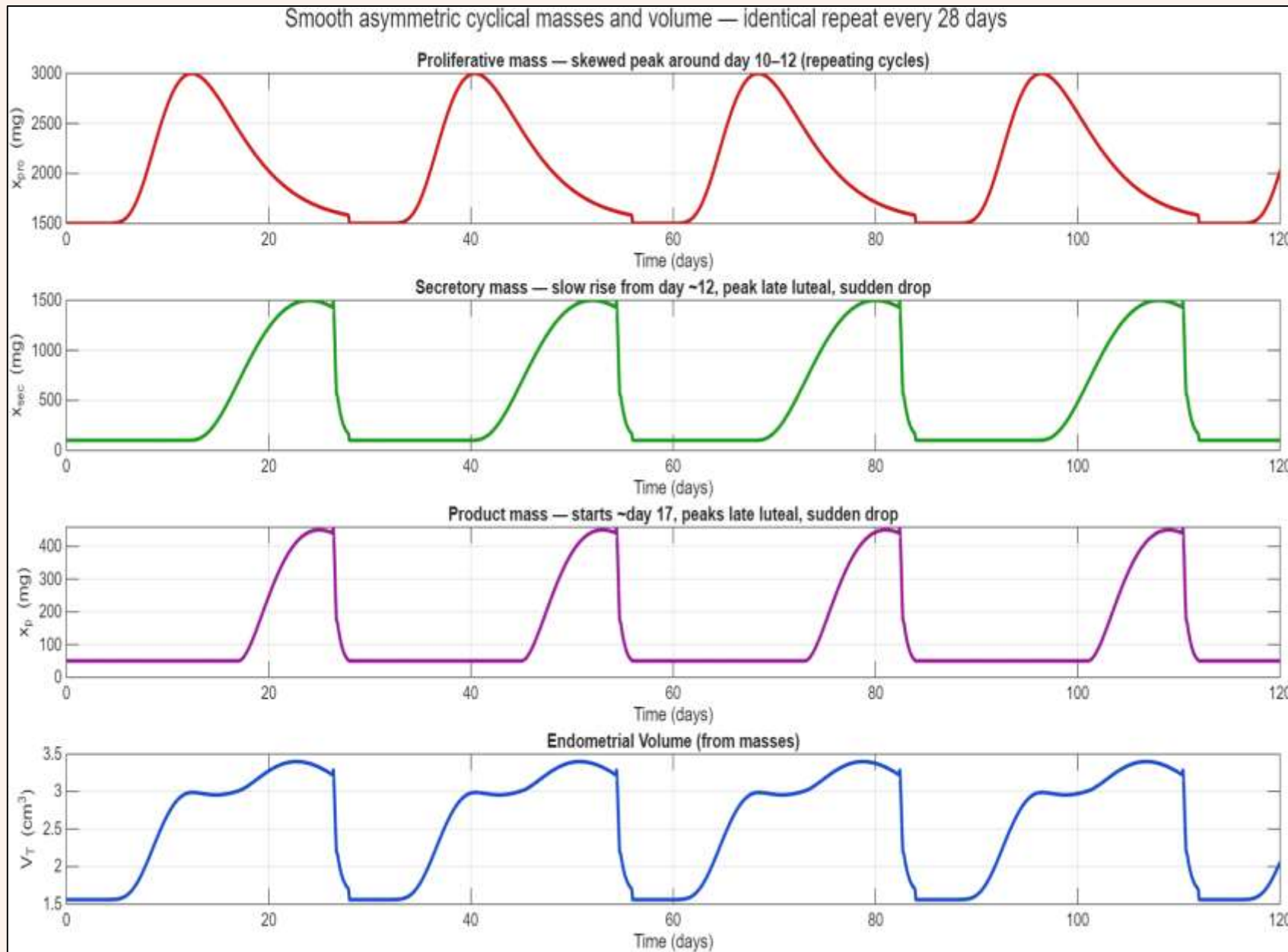
|        |                          |
|--------|--------------------------|
| $x$    | Cellular mass            |
| $\mu$  | Specific Growth Rate     |
| $S_B$  | Enzymatic processes      |
| $w$    | Mass function            |
| $Y$    | Yield Factor             |
| $\rho$ | Density of cells         |
| $V_T$  | Total Endometrial Volume |

Cyclic Endometrial Dynamics — Estrogen, Progesterone, and Volume



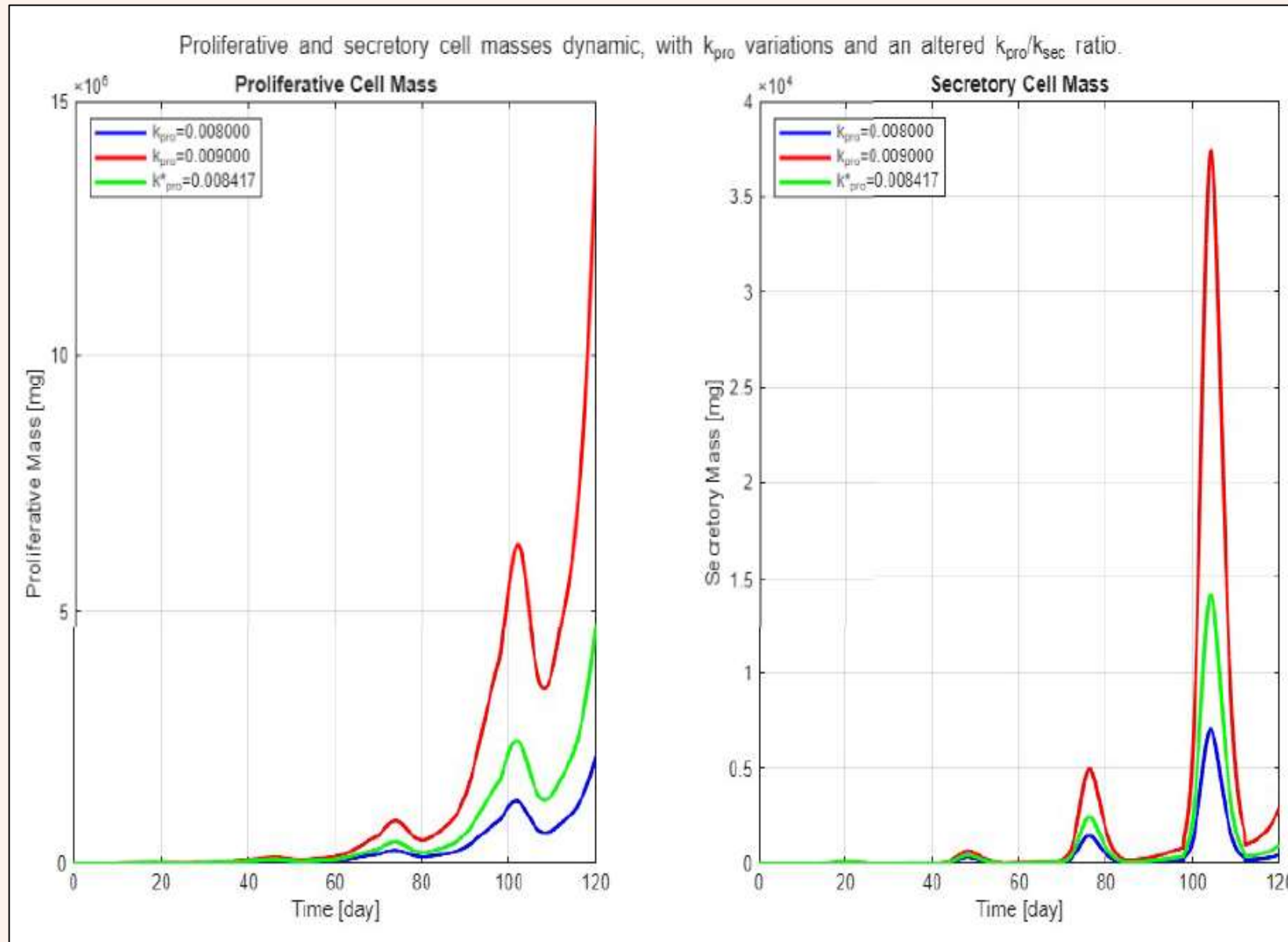
## Graphs

The hormones show oscillatory pattern, due to their periodicity, the endometrial thickness also periodically increases and decreases showing direct relationship between hormonal levels and endometrial physiology.



## Graphs

Cellular masses vary periodically with time over a 30 days cycle and therefore endometrial volume varies proportionately.

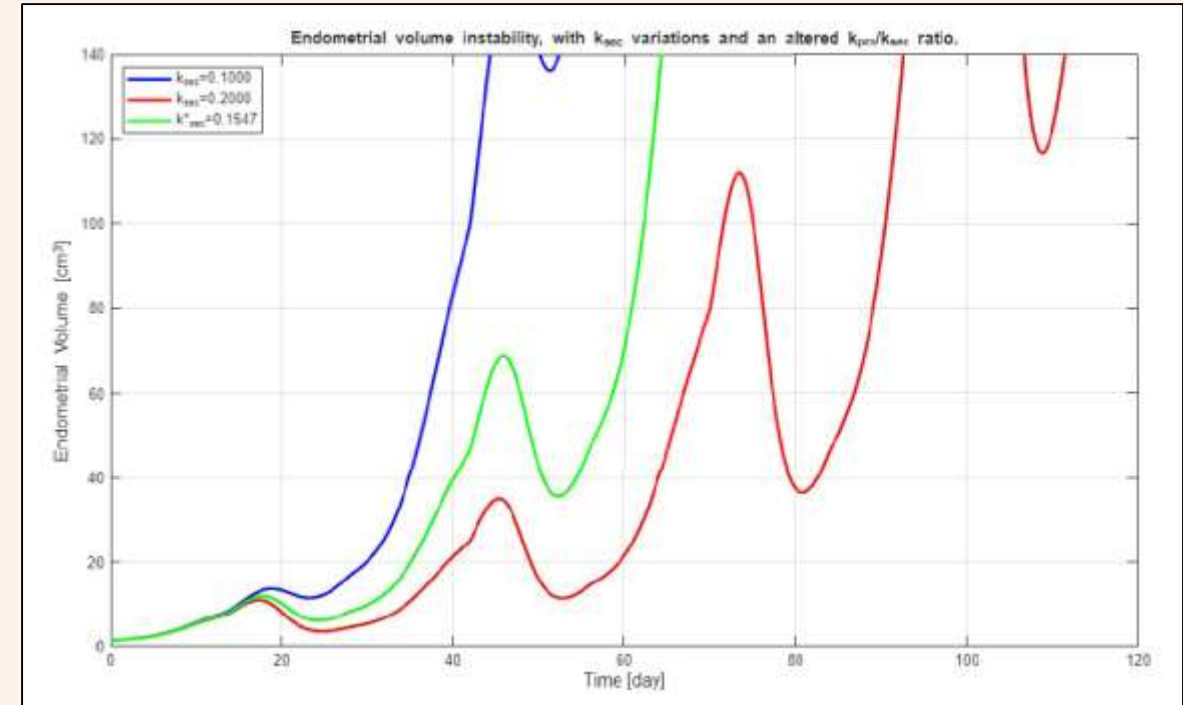
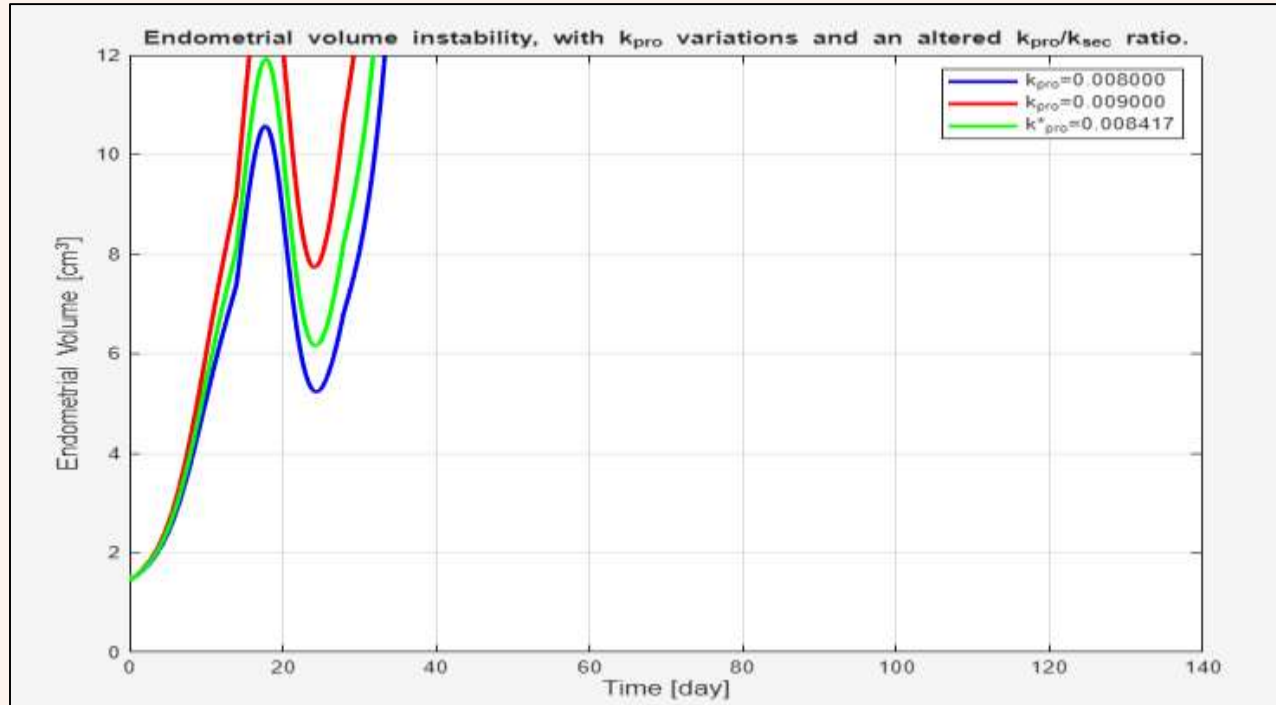


## Graphs

When the values of  $k_{pro}$  were varied keeping  $k_{sec}$  constant, altering their ratio, the plots of both proliferative mass and secretory mass show increase in cellular mass over time, indicator of disease.

# Graphs

These plots again show that endometrial volume instability is observed when proliferative and secretory cell mass ratio is altered.



# Conclusions

- Estrogen and Progesterone both play significant role in the progression of endometriosis.
- When the balance between the two is altered, rapid growth in cellular mass is observed.
- When the ratio remains constant, but the levels of both hormones increase proportionally, endometrial volume does not show significantly dynamic behavior.
- Increase in either of the two hormones shows increase in endometrial volume suggesting abnormality in uterus which can lead to endometriosis.

# Future Directions

- The model clearly points towards the significance of balance between estrogen and progesterone in the progression of endometriosis therefore, experimentalists should also explore progesterone dynamics in the disease development so that the pathology can be better understood.
- The model can be modified with the addition of the role of other hormones like inhibin h and amh as it is known that inhibin inhibits the gonadotropin release from the pituitary along with P4 and E2 and amh has role in menstruation.
- The model has only considered for time scales of a few months, the dynamics could be studied for longer period since endometriosis is usually detected after years when the progression is high, understanding the longer dynamics would help us in exploring smaller cues that point to potential pathology.

Thank You!